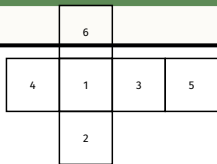
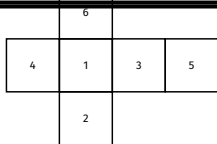


Push Tasks - Nets

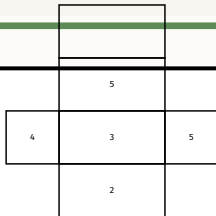


1. Invalid arrangements?

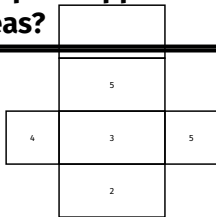
e.g. 4 in a row with 2 on



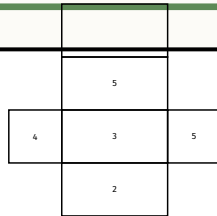
2. Why some cross shapes fail?



3. 3 pairs opposite face areas?

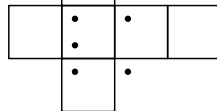


4. Min SA for $V=96$?



5. Max SA for $V=96$?

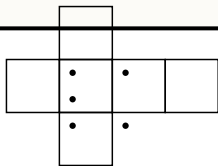
1x1x96 gives SA=296



6. Opposite faces sum to 7 proof?

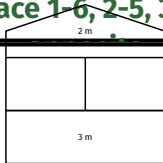
Standard die: 1-6 2-5

Push Tasks - Nets



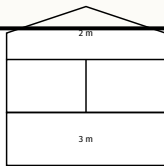
7. Design die net?

Place 1-6, 2-5, 3-4



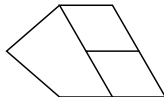
8. Triangular faces isosceles?

Two sides equal by



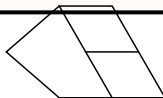
9. Different tent net?

e.g. swap positions of



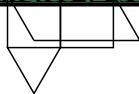
10. Triangles equal?

Yes — same base and height



11. Rectangular base pyramid?

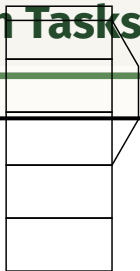
Net has 1 rectangle + 4 triangles (2 pairs)



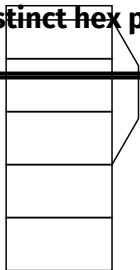
12. Right triangle prism net?

Rectangles + 2 right triangles

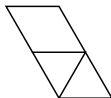
Push Tasks - Nets



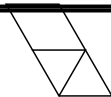
13. Distinct hex prism nets?



14. Hex prism SA?

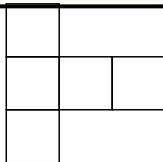


15. Tetrahedron nets?
2 distinct nets

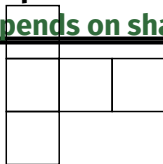


16. 4 triangles needed?

Tetrahedron has exactly 4 faces



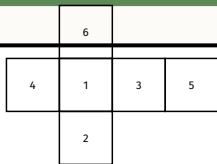
17. Lid positions count?
Depends on shape



18. $V=12$ missing top SA?

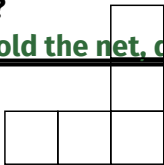
Depends on

Push Tasks - Nets



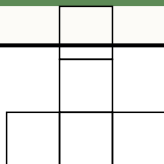
19. Shortest surface path?

Unfold the net, draw



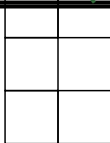
20. L-shape cube net condition?

The L must have 6

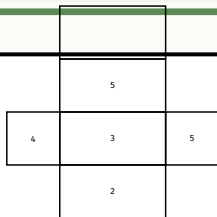


21. 2 non-box pentominoes?

1-pentomino (straight

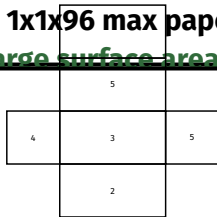


**22. All open box nets?
11 pentominoes can**



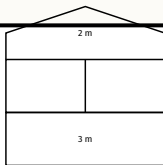
23. 1x1x96 max paper?

Large surface area —



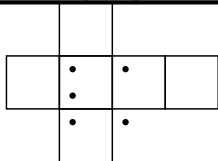
24. 4x6x4 min paper proof?

Push Tasks - Nets



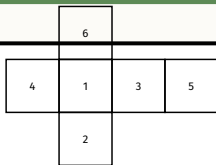
25. Ribbon on faces?

3.8 m on faces, 0.2 m



26. Minimal colours?

3 colours needed



27. Shortest path on cube?

**Draw net, connect
start to end**